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CSA06 - Design and Analysis of Algorithms

DESIGN AND ANALYSIS OF ALGORITHMS

ASSESSMENT TOOL 1 – Coding Challenge Task

Q23. MINIMUM NUMBER OF JUMPS

Given an array where each element represents the maximum number of steps that can be taken forward, determine the minimum number of jumps required to reach the last index. The goal is to minimize the number of moves while ensuring reachability. Use Dynamic Programming or Greedy techniques to compute the optimal solution.

SAMPLE INPUT:

n=5
arr = 2 3 1 1 4

EXPECTED OUTPUT:

Min Jumps = 2

AIM

To find the minimum number of jumps required to reach the last index of an array, where each element represents the maximum number of steps that can be taken from that position.

ALGORITHM

1. Read the array elements.
2. Initialize jumps = 0, currentEnd = 0, and farthest = 0.
3. Traverse the array and calculate the farthest reachable index.

4. When the current reachable range ends, increase jumps and update currentEnd.
5. Continue until the last index is reached.
6. Display the minimum number of jumps.

PSEUDOCODE

```
START
Read n
Read array arr[0...n-1]
If n <= 1
   Print "Min Jumps = 0"
   STOP
jumps = 0
currentEnd = 0
farthest = 0
For i = 0 to n-2
   farthest = max(farthest, i + arr[i])
   If i == currentEnd
      jumps = jumps + 1
      currentEnd = farthest
      If currentEnd >= n-1
         BREAK
If currentEnd < n-1
   Print "Last index cannot be reached"
Else
   Print jumps
STOP
```

MANUAL PROCESS

Given:

Array = 2 3 1 1 4

Index = 0 1 2 3 4

Step 1

Start at index 0.

arr[0] = 2

From index 0, we can reach index 1 or 2.

0 → 1 or 2

So the first jump can reach up to index 2.

Step 2

Check the reachable positions.

At index 1:

1 + arr[1]

= 1 + 3

= 4

So index 1 can directly reach the last index 4.

Therefore: 0 → 1 → 4

Step 3

Number of jumps:

0 → 1 = Jump 1

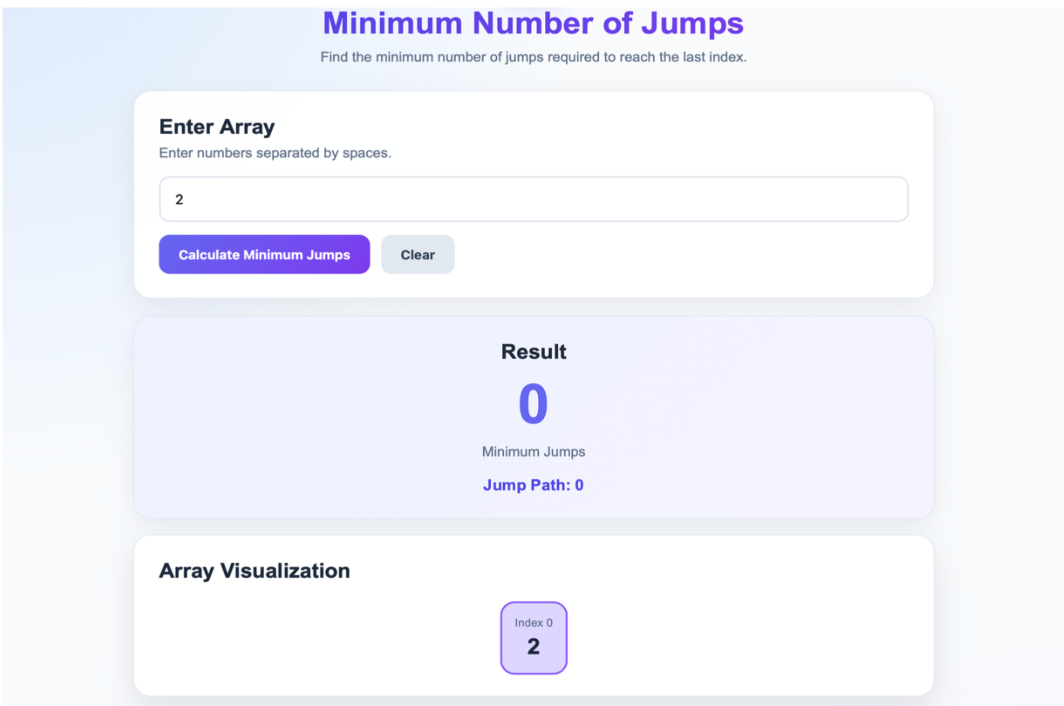
1 → 4 = Jump 2

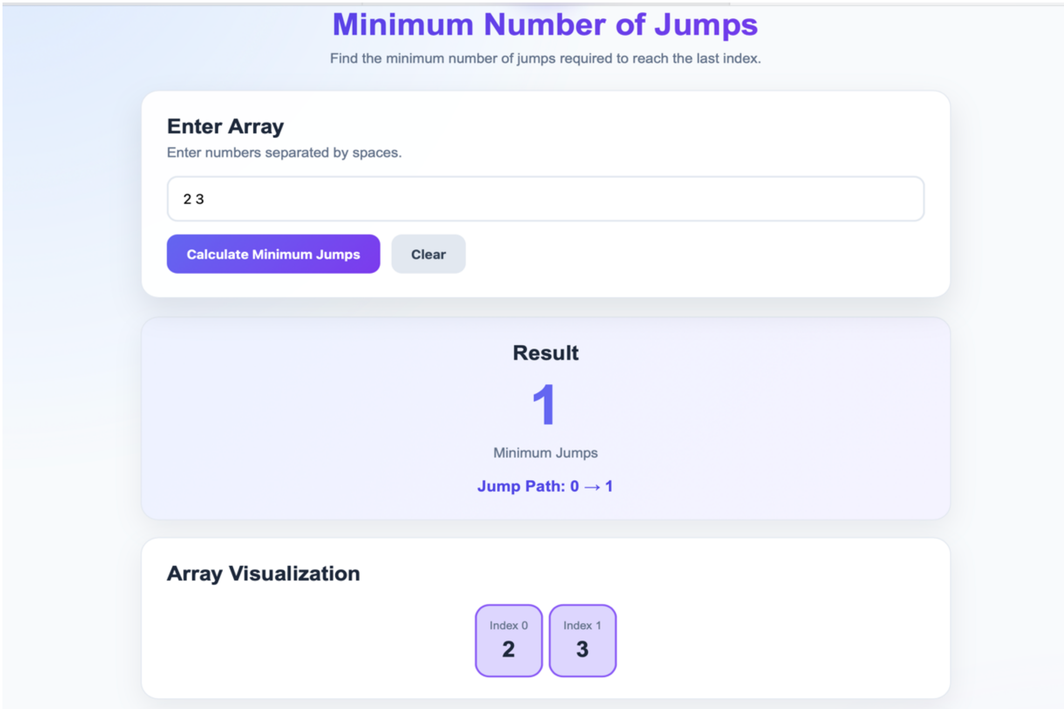
Therefore:

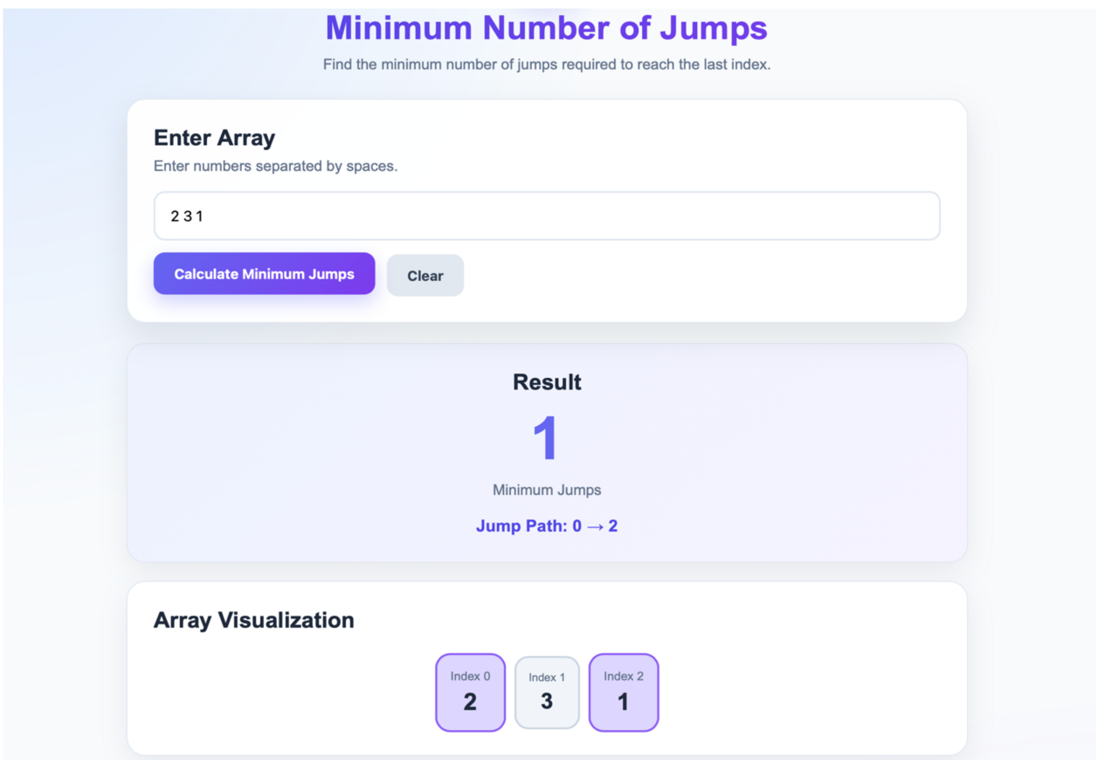
Min Jumps = 2

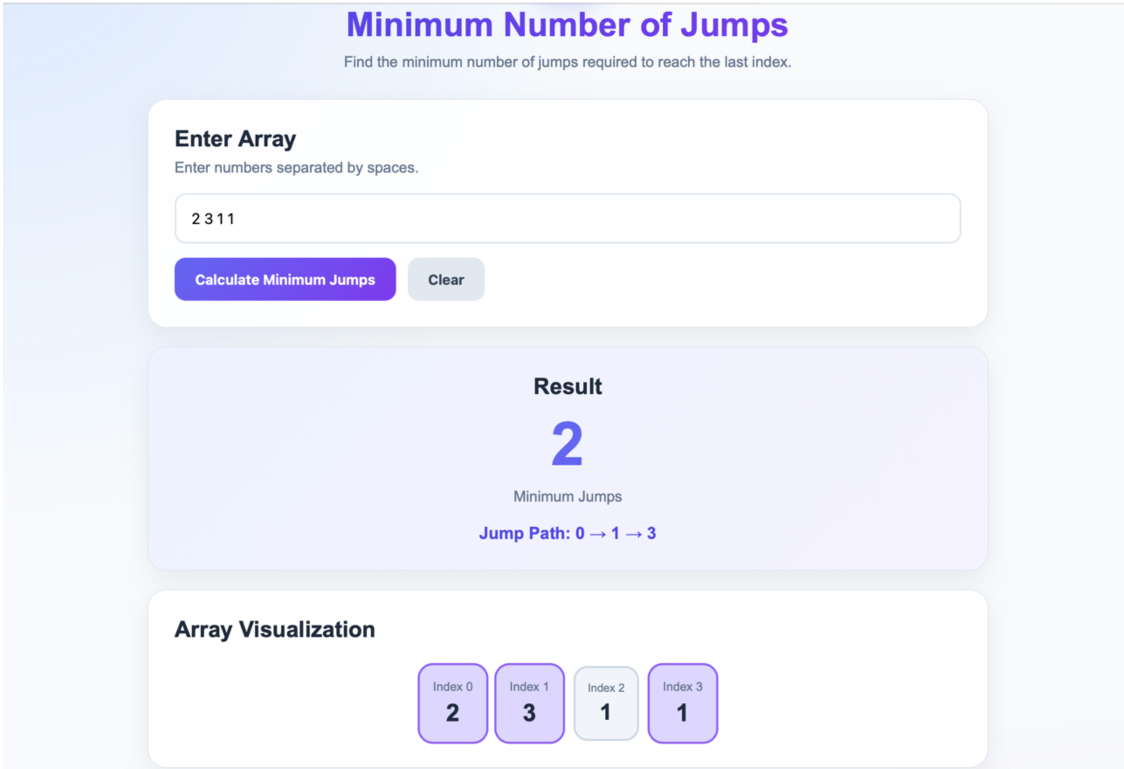
RESULT

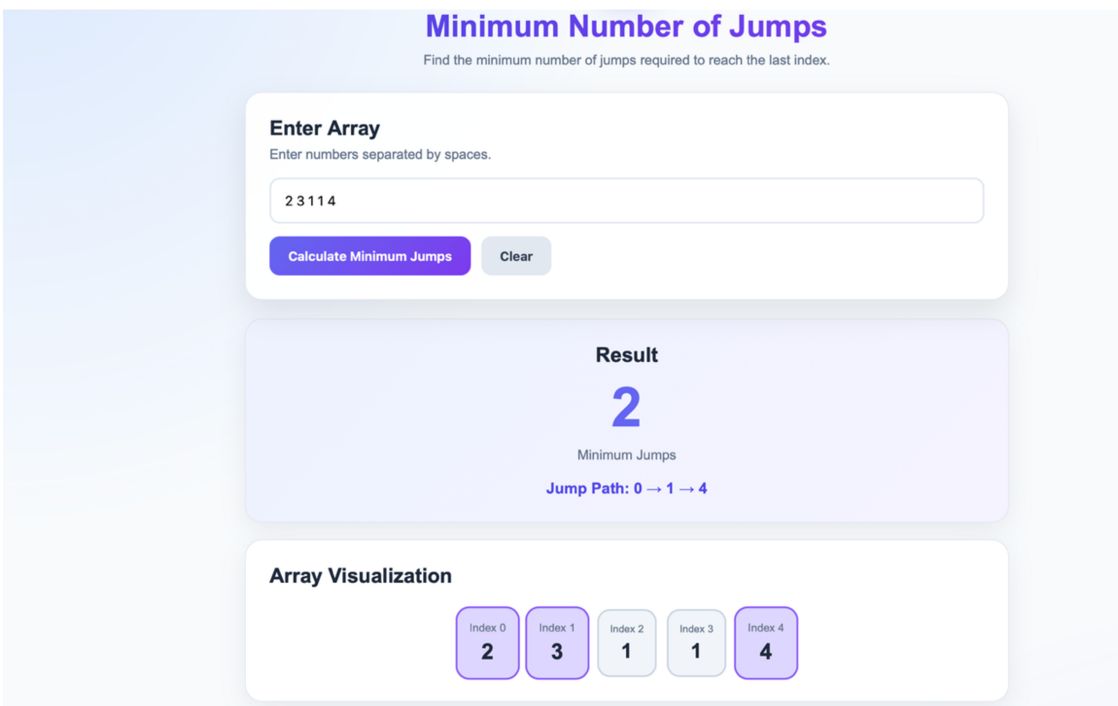
The minimum number of jumps required to reach the last index is 2.




Minimum Number of Jumps
Find the minimum number of jumps required to reach the last index.
Enter Array
Enter numbers separated by spaces.
2 3
Calculate Minimum Jumps
Clear
Result
1
Minimum Jumps
Jump Path: 0 → 1
Array Visualization
Index 0
2
Index 1
3


Minimum Number of Jumps
Find the minimum number of jumps required to reach the last index.
Enter Array
Enter numbers separated by spaces.
2 3 1
Calculate Minimum Jumps
Clear
Result
1
Minimum Jumps
Jump Path: 0 → 2
Array Visualization
Index 0
2
Index 1
3
Index 2
1





CONCLUSION

Thus, the minimum number of jumps required to reach the last index is successfully calculated using the Greedy approach. For the given input 2 3 1 1 4, the minimum number of jumps is 2**.**